$CO_1 - AT_1$.

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.

A. Defination :-

cloud computing is the delivery of computing services such as servers, storage, databases, network and software area the internet instead of maintaining physical infrastructure on-premises four characteristics.

Four characteristics :-

1. on-demand self-service :- user can access computing resources whenever required without directly contacting the service provider.

2. Broad network access :- cloud services are accessible through the internet using laptops and other devices.

3. Resource pooling :- computing resources can be shared among multiple users.

4. Rapid elasticity :-

Resources can be increased on decreased quickly according to demand.

2. Trace the evolution of cloud computing from hardware evolution.

cloud computing developed through serval technological

1. Hardware Evolution :- Powerful computers and servers made it possible to process large amounts of data efficiently.

2. Distributed Computing : multiple computers worked together to share workloads and computing resources

3. virtualization :- A single physical server cloud run multiple virtual machines, improving resources utilization and reducing hardware cost.

4. Internet Evolution : Faster and more reliable internet connections enabled users to access computing resources remotely.

3. server virtualilization: It is a technology that divides one physical server into multiple virtual machines. Each virtual machine run its own operating system and applications.

Roles of virtualization in cloud computing :-

1. Improving hardware utilization by running multiple virtual machines on one physical server into multiple virtually server.

2. Reduces hardware and operational cost

3. Supports flexible and scalable cloud infrastructure

Virtualization	cloud Computing
* It is an enabling technology	* It is service delivery model.
* creats virtual machines from physical hardware	* Delivery computing resources over internet
* mainly improves hardware utilization Eg:- VMware	* provides on-demand access to computing services. Eg:- Amazon web servies.

Services model	Full form	Real-world example
Iaas	Infrastructure as a service	Amazon EC2
Paas	Platform as a service	Google app engines
Saas	Software as a service	Gmail
Xaas	Anything as a service	DropBox / Storage as a service

Explanation :—

Iaas :- users manage their operating system and applications, while the provider manges the underlying infrastructure.

Paas :- Developers focus on building applications without managing the underlying servers.

Saas :- user access complete software application through a browser or app.

Xaas :- A broad term covering services such as storage, security, networking and software delivered over the internet.

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